

|                              |        |               |                  |                  |              |           |        |          |                     |              |                         |                      |           |                         |           |         |    |
|------------------------------|--------|---------------|------------------|------------------|--------------|-----------|--------|----------|---------------------|--------------|-------------------------|----------------------|-----------|-------------------------|-----------|---------|----|
| Varen 6869 AgSp1.1 ct        | NVGS   | SD            | SGERTVNQLVQSV    | ISGINIPGNYVDYNPF | FS           | ELSSQLS   | QLRTGT | .DKPTKQ  | .LISVLMEGLIAALEALNA | AKISGFRSDV   | FVSSDLPVYI              | 93                   |           |                         |           |         |    |
| Varen 6869 AgSp1.2 cta       | NVDSSV | GEQAVSKLMKAL  | ISGINIPGNYVDYN   | AF               | FS           | ELSSQLYGM | QTGT   | TSERPTKQ | .FMSVLMEGLIAAMEALNA | AKISGFRNDVSV | SGDLPVYT                | 94                   |           |                         |           |         |    |
| Varen 6869 AgSp1.2 ctb       | NVGSSA | GEQKVSQMLMQSL | ISGINISGNYVDYND  | FF               | FS           | ELSSQLSGM | RTV    | TT       | ERPT                | EQ           | .LISVLMEGLIAAMEALNA     | AKISGFRNDVSVSGDMPVYT | 94        |                         |           |         |    |
| Mg TRINITY DN1546 c0 g1 Ct 1 | NVDSRD | GKENINQLMQAL  | ISGINISKNYVDYSS  | FF               | RELSFQLSQL   | GTGETKR   | PTKQ   | Q        | IISVLMEGLIAGLET     | LNSAKISGFRND | ISVSV                   | DLPVYA               | 95        |                         |           |         |    |
| Mg TRINITY DN1546 c0 g1 Ct 2 | NVDSRT | GKTVNQLMH     | ALISGISIPENYVHYN | VFF              | GE           | ELSSQLSEL | RTGT   | AE       | RP                  | TKQ          | .LISVLMEGLIAGLET        | LNAKISGFRNDVSVAGDLP  | MYT       | 94                      |           |         |    |
| Mruby 7048 AgSp1 ct          | DVESS  | TGEQTVS       | QLMQAII          | ISGINIPGNYVDYND  | FF           | NQLSS     | LLS    | QVRQS    | .ADSPT              | EQ           | .VDSVLMEALIAALEALNA     | AAKVS                | GFRNDVSI  | IGSDLPVYT               | 93        |         |    |
| Ncruc 58532and13532 AgSp1 ct | NVESSV | GQQT          | VSQMLQAI         | ISGINIPGNYVDYND  | FF           | NQLSS     | LLS    | QVRSG    | .TDSPT              | TKQ          | .VDSVLMEALIAALEALNSAKVT | GFRNDVSV             | ANDLPVYT  | 93                      |           |         |    |
| Msag 18528 AgSp1.2 ct        | TVGSSS | GAQTVKQLKQAI  | VSGINIAGNYVDYNG  | FF               | RELSAQLS     | NIQT      | GETAK  | PTKE     | .LISTLMEGLIAALEALNA | AKISGFRT     | .VSVTSDLPVYT            |                      | 93        |                         |           |         |    |
| Lcorn 51676 AgSp1.1Cfrag ct  | NVRSSA | GEQTVS        | QLMQAII          | ISGINIPGNSVDYND  | FF           | NQLSS     | LLS    | QVRSG    | STDSPT              | TKQ          | .VDSILMEALIAALEALNSAKVT | GFRNDVSI             | IGSDLPVYT | 94                      |           |         |    |
| Lcorn 58650 AgSp1.2 ct       | NVRSSA | GEQTVS        | QLMQAII          | ISGINIPGNSVDYND  | FF           | NQLSS     | LLS    | QVRSG    | STDSPT              | TKQ          | .VDSILMEALIAALEALNSAKVT | GFRNDVSI             | IVSDLPVYT | 94                      |           |         |    |
| Amarm 50219 AgSp1.1 ct       | SVKSS  | TGEQTVS       | QLMQAII          | ISGINIPGNYVDYND  | FF           | NQLSS     | MLS    | QVRSG    | STDSPT              | TKQ          | .VDSILMEALIAALEALNSAKVT | GFRDDVSV             | ASDLPVYA  | 94                      |           |         |    |
| Aarg AgSp1 FC511 ct          | NVASNA | GEQSVS        | QLMQAII          | ISGINISGNYVDYND  | FF           | NQLSS     | QLS    | QVR      | TGPAES              | PTKQ         | .LMEVLMEGLIAALEALNSAKVT | GFRNDVSI             | ISSDLPVYT | 94                      |           |         |    |
| Atri AgSp1 genomic ct        | NVASST | GETVS         | QLLHDI           | ISGINISGNYVDYND  | FF           | NQLSS     | LLS    | QVR      | TDPTD               | SPSKQ        | .LIKILMEGLIAALEALNSAKVT | GFRSDVSI             | IDS       | DLPVYT                  | 94        |         |    |
| Mhut 1271 AgSp1 ct           | NVESS  | TGEQTVS       | QLMHS            | II               | ISGINIPGNYVD | FNDFF     | NQ     | ISS      | LLS                 | QVRSG        | STNS                    | PT                   | EQ        | .VDSILMEALIAALEALNSAKVT | GFRKDVSVA | SDLPVYT | 94 |

|                              |                                          |     |
|------------------------------|------------------------------------------|-----|
| Varen 6869 AgSp1.1 ct        | SFLNDVL                                  | 100 |
| Varen 6869 AgSp1.2 cta       | IFLNEILEMPST.TPVSPKDGTCADDVASIFSTAKQALR  | 132 |
| Varen 6869 AgSp1.2 ctb       | TFFNEIFY                                 | 102 |
| Mg TRINITY DN1546 c0 g1 Ct 1 | TFLNEVFDMYSPTFSVSVSKDENCNGNLGSTFGTVKQALQ | 134 |
| Mg TRINITY DN1546 c0 g1 Ct 2 | AFLNKI                                   | 100 |
| Mruby 7048 AgSp1 ct          | SFLNEILY                                 | 101 |
| Ncruc 58532and13532 AgSp1 ct | SFLNEILY                                 | 101 |
| Msag 18528 AgSp1.2 ct        | SFLNEILF                                 | 101 |
| Lcorn 51676 AgSp1.1Cfrag ct  | SFLNEILY                                 | 102 |
| Lcorn 58650 AgSp1.2 ct       | SFLNEILY                                 | 102 |
| Amarm 50219 AgSp1.1 ct       | SFLKEILY                                 | 102 |
| Aarg AgSp1 FC511 ct          | SFLNEIL                                  | 101 |
| Atri AgSp1 genomic ct        | SFLNEILY                                 | 102 |
| Mhut 1271 AgSp1 ct           | SFLNEIFF                                 | 102 |

X non-conserved  
X similar  
X ≥ 50% conserved